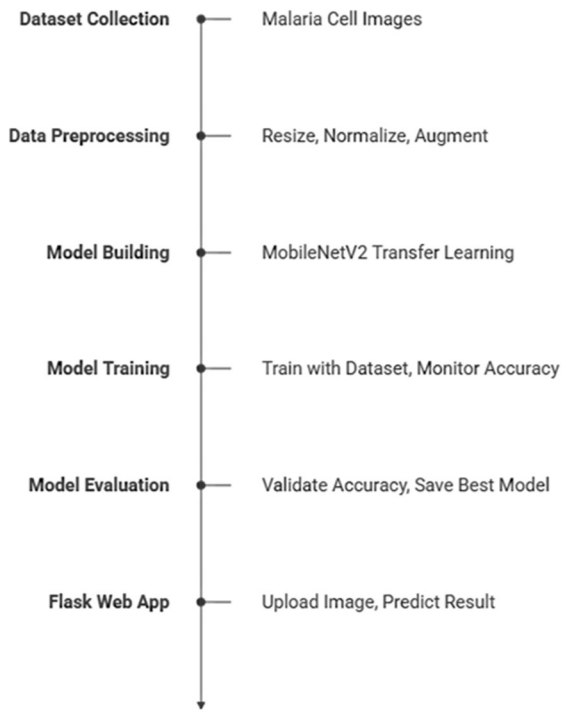
AI Based Malaria Detection Using Deep Learning

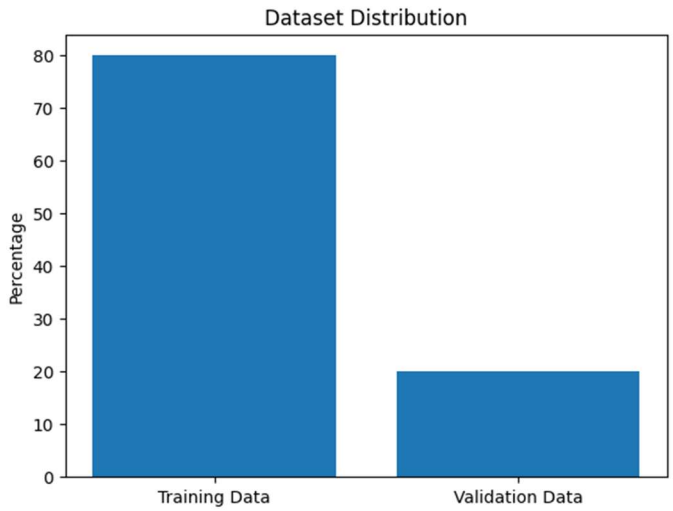
PROJECT FLOW

Introduction

Malaria is a life-threatening disease caused by parasites transmitted through infected mosquito bites. Traditional malaria diagnosis involves examining blood smear images under a microscope, which requires expert knowledge and can be time consuming. Deep learning techniques can assist medical professionals by automatically detecting malaria parasites from microscopic images.
In this project, a deep learning model based on MobileNetV2 is used to classify blood cell images as infected or uninfected. The trained model is integrated into a Flask web application that allows users to upload images and receive predictions.

System Workflow





Data Collection and Preprocessing

The malaria dataset contains microscopic images of infected and uninfected blood cells. The dataset is organized into separate folders for each class. Images are resized and normalized before being used for training. Image augmentation techniques such as rotation and flipping help improve model generalization.

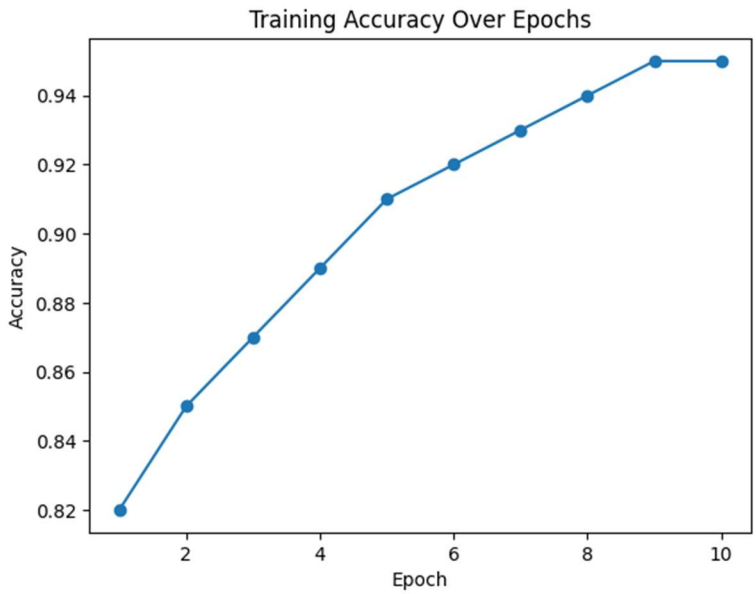
Model Building

The MobileNetV2 architecture is used with transfer learning. The pre trained base model is loaded and custom classification layers are added on top. The model is compiled using the Adam optimizer, binary cross entropy loss, and accuracy as the evaluation metric.

Model Training

The model is trained using image data generators that load images in batches. Training accuracy and validation accuracy are monitored across multiple epochs. Early stopping and learning rate reduction are used to improve training performance.

Training Performance



Application Development

A web application is developed using Flask. Users can upload a microscopic blood smear image through the web interface. The backend preprocesses the image and uses the trained deep learning model to predict whether the cell is infected or uninfected. The prediction result is then displayed on the webpage.

Conclusion

This project demonstrates how deep learning can assist in malaria detection using image classification techniques. By integrating the trained model with a web application, the system provides a simple and efficient method for malaria diagnosis.